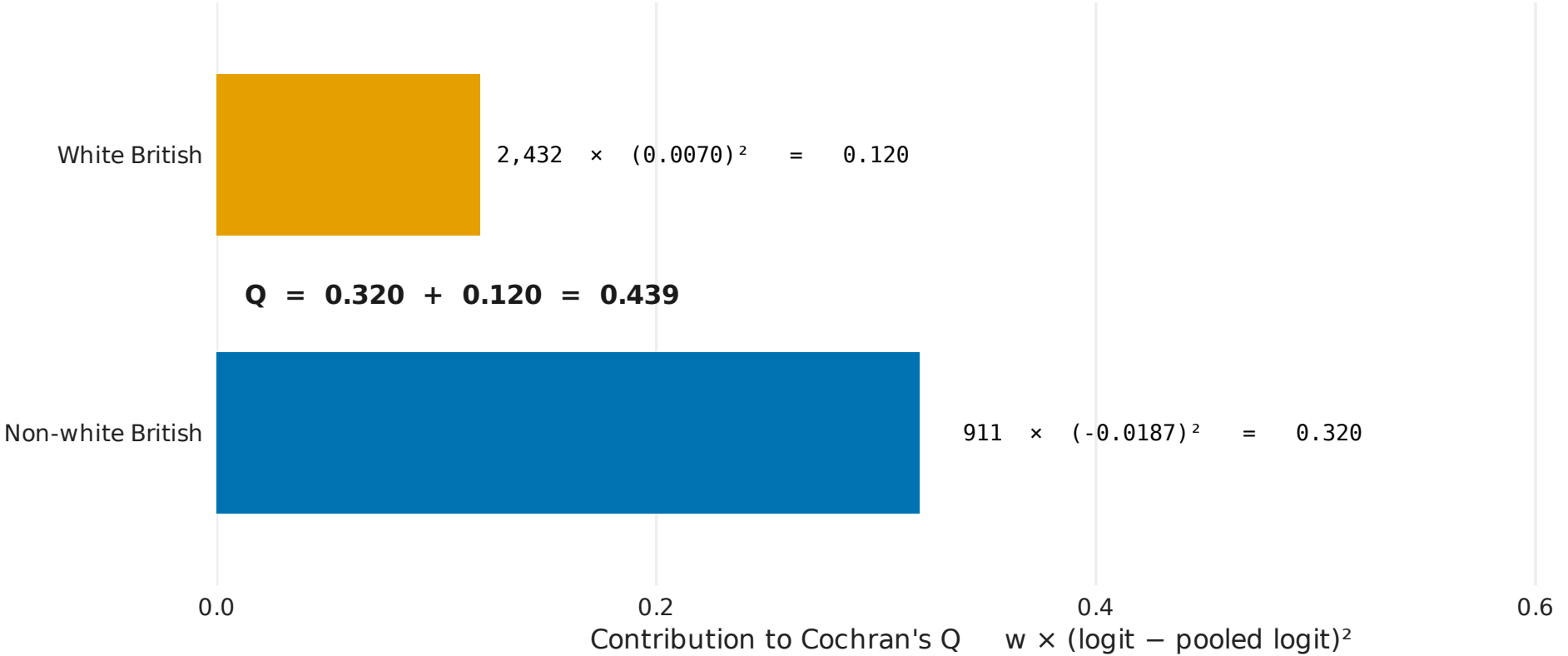
4. Cochran's Q is the sum of the weighted squared distances

Square each distance from the balance point in plot 3, so the sign drops out.
Multiply by that study's weight, then add the two products. That sum is Q.



Q = 0.439. It describes scatter around the pooled logit. It does not replace that logit, and the pooled AUC stays 0.809.
The less precise study contributes more (0.320 versus 0.120) because it sits further from the mean, and squaring magnifies that gap.